Filtering Rows with WHERE

Lecture Notes · Game DB Junior

In this lesson

- How to **filter** rows with the **WHERE** keyword
- The **order** in which Postgres runs FROM, WHERE and SELECT
- Why this order helps us write harder queries later

1. Why filter?

Until now, every query we wrote gave us **all the rows** in the cities table. But often we only want the rows that **match some criteria**. In other words, we want to **filter** the result set.

Example: we do not want every city. We only want the cities with an **area greater than 4000**.

2. The WHERE keyword

After the FROM part, we add the filter. We use the keyword **WHERE**, and right after it we write our criteria:

```
SELECT name, area FROM cities WHERE area > 4000;
```

- **WHERE** — the keyword that starts the filter.
- **area > 4000** — the criteria: area greater than 4000.
- When we run it, we only get **two results**: Tokyo and Shanghai. Both have an area greater than 4000.

3. How Postgres runs this query

Having a good picture in your head of how this query works is **really key** for writing more complicated queries later.

Do not read it from left to right. You may think Postgres runs the query in the same order we write it: SELECT, then FROM, then WHERE. It is best **not** to think of it this way.

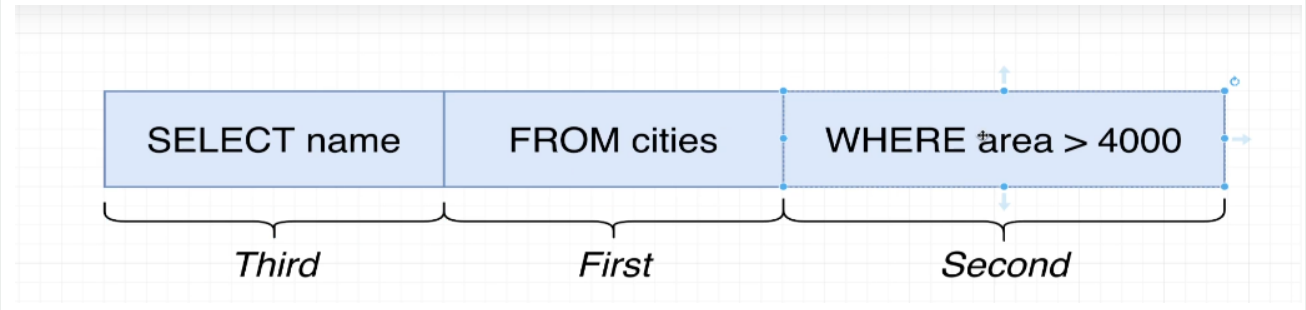


Figure 1 — The order: FROM first, WHERE second, SELECT third

Order	Part	What happens
First	FROM cities	Postgres looks at the data source. It gets all the rows from the cities table
Second	WHERE area > 4000	It applies the filter to each row. Only rows with an area greater than 4000 stay. The other rows are removed
Third	SELECT name, area	From the rows that are left, it picks the columns we asked for

Step by step with our data

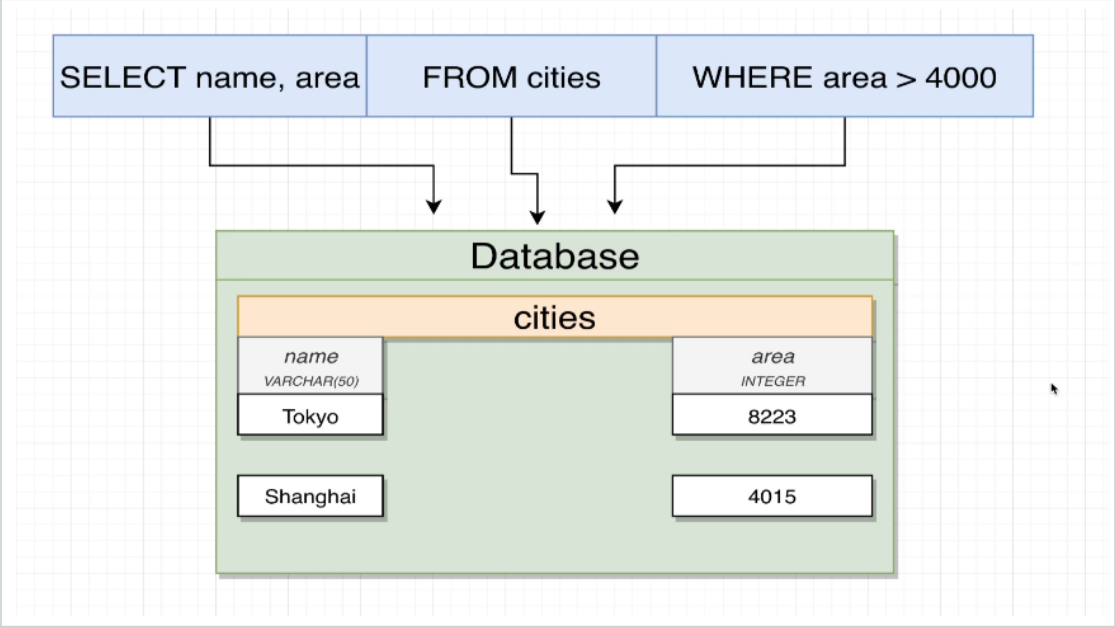


Figure 2 — The result after FROM, WHERE and SELECT

- **FROM** gives us all the rows in cities: Tokyo, Delhi, Shanghai, Sao Paulo.
- **WHERE** removes Delhi and Sao Paulo, because their area is less than 4000. Now only Tokyo and Shanghai are left.
- **SELECT** takes only the **name** and **area** columns from these rows. This is our output.

We will use this idea of **which order the keywords run in** again and again. When you can picture the whole process, writing harder queries becomes much easier.

4. Running the code (my practice)

I wrote the query with a short comment and ran it.

```sql
-- Show the name and area of cities with an area greater than 4000
SELECT
    name, area
FROM cities
WHERE area > 4000;
```

Figure 3 — SELECT with WHERE area > 4000

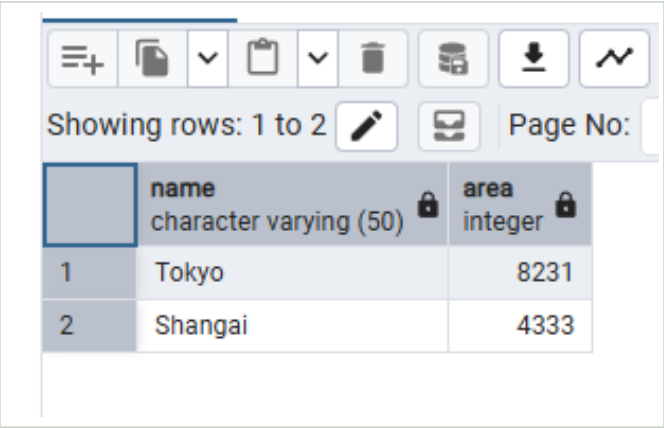


Figure 4 — Result: only Tokyo and Shangai, 2 rows

5. Summary

Idea	Remember
Filter	Get only the rows that match some criteria
WHERE	Keyword for the filter. Comes after FROM. The criteria comes right after it
Order of running	FROM first → WHERE second → SELECT third
Not left to right	Do not think of the query as SELECT → FROM → WHERE

There are many other ways to use WHERE. We will see more examples in the next lesson.

Sources
Lecture transcript and lecture images (Figures 1–2)
My own practice screenshots (Figures 3–4)